VECTORS I TEST
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Higher 2

MATHEMATICS 9758

Vectors I **2026**
QUESTION PAPER

Additional Materials: Writing paper
List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are **not** allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 42.

Full sourced questions. Give exact answers unless otherwise stated.

1 With reference to the origin O, the points A and B have coordinates $(1, 2, 0)$ and $(-5, 1, 0)$.

(a) Find the angle AOB. [3]

The vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are projected along the direction of $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ to form the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$ respectively.

(b) Find the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$. [3]

(c) Find the exact area of the triangle $O'A'B'$. [2]

[DHS 2023 Promo Q5]

2 The position vectors of points A, B and C, relative to the origin O are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}, \quad \mathbf{b} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}, \quad \mathbf{c} = 3\mathbf{i} - \mathbf{j} - \mathbf{k}.$$

(a) Find $\mathbf{a} \times \mathbf{b}$ and deduce the exact area of triangle OAB. [3]

(b) Hence, find the volume of the tetrahedron $OABC$.
[The volume of a tetrahedron is $\frac{1}{3} \times$ base area $\times$ perpendicular height.] [3]

(c) Find the perpendicular distance from B to line OA. [2]

[NYJC 2024 Promo Q5]

3 Referred to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. C lies on AB such that $AC : AB = 1 : 3$.

(i) Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the area of triangle OAC is given by $k|\mathbf{a} \times \mathbf{b}|$, where k is a constant to be determined. [2]

It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

(iii) Using the result in part (i), find the value of $(2\mathbf{a} - \mathbf{b}) \cdot \mathbf{c}$. [3]

[MI 2024 Promo Q4]

4 **(a)** The points A, B and C have position vectors $\begin{pmatrix} 1 \\ 3 \\ -7 \end{pmatrix}$, $\begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -9 \\ 8 \\ -27 \end{pmatrix}$, with respect to the origin O.

(i) Show that the points A, B and C are collinear. [2]

(ii) Find the exact area of the triangle OAB. [3]

(b) The vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are defined such that $|\mathbf{u}| = 1$, $|\mathbf{v}| = 2$ and $|\mathbf{w}| = 3$.

(i) Suppose that $\mathbf{u} + \mathbf{v} = \mathbf{w}$. With the aid of a diagram, or otherwise, explain why $\mathbf{u}$ is a unit vector that is parallel to $\mathbf{w}$. [2]

(ii) Given instead that each of the three vectors is perpendicular to the sum of the other two vectors, find the exact value of $|\mathbf{u} + \mathbf{v} + \mathbf{w}|$. [4]

[TMJC 2024 Promo Q6]

5 Relative to the origin O, the points A and C have position vectors $\mathbf{a}$ and $2\mathbf{b}$ such that

$$\mathbf{a} = 5p\mathbf{i} - 2p\mathbf{j} + 4p\mathbf{k} \quad \text{and} \quad \mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k},$$

where p is a positive constant. It is given that $|\mathbf{a}| = 2|\mathbf{b}|$.

(a) Find the exact value of p. [2]

(b) Evaluate $(\mathbf{a} + 2\mathbf{b}) \cdot (\mathbf{a} - 2\mathbf{b})$. [1]

(c) Evaluate $\mathbf{a} \times \mathbf{b}$ and hence find the area of triangle OAC. [3]

(d) Use a geometrical reason to explain why $|\mathbf{a} \times \mathbf{b}| = \frac{1}{4}|\mathbf{a} + 2\mathbf{b}||\mathbf{a} - 2\mathbf{b}|$. [2]

[RI 2025 Prelim P1 Q3]

END OF TEST